Appendix A - Prototype Fabrication Instructions

How to build the flywheel launcher - A Manufacturer's guide

The flywheel launcher system was designed to be easily taken apart for maintenance and just as easily put together. A full instruction set, parts list, and tools list are provided in the following manual for manufacturers contracted by the RHS Cornerstone. The construction of the prototype takes no more than 2 hours per unit, assuming all parts and tools are obtained and laid out prior to beginning.

Tools List

Red Robertson screwdriver
Needle-nose pliers
Wire stripper pliers
Allan key set
7/16in. Box/open-end wrench
1/8in. Terminal driver
Flush cut pliers
Phillip's screwdriver
Power Drill
0.25in. drill bit
0.125in. drill bit
Multi-tool/router (semi-precise cutting tool)
Measuring utensil of some sort (a scale or caliper for best accuracy)
Marking tool (Sharpie)

Parts List (Dimensions given in “L x W x H”. If circular “Diameter x length”

Trailer:
1 – 14in. x 6in. x 3in. dump-style trailer
2 – 4.3in. x 0.4in. Stainless steel plates

Firing/reload system:
1 - 18-volt, 2 amp-hour battery
2 - 18-volt DC motors with 30mm x 2mm output shafts
2 – 0.25in. x 1.1in. flywheels with 2mm receivers
1 – MG90 2KG*cm continuous servo motor

1 - 0.5in. x 4in. x 0.25in. piece of stainless steel
1- 1.1in. x 4 in. Plastic cylinder
8.75in. of 3mm diameter brass rod

Barrel:
2 feet 1.5in. PVC
1 – 1.5in. 3 – way PVC connector

Electrical Hardware:
1 – 5.5in. x 2.1in. Breadboard
1 – Arduino Uno smartboard
1 – XY160D Power supply module
10 – 50cm female to male jumper wires
6 – 12cm male to male jumper wires
2 feet of both green and black 16-G wire
4 - plier crimp terminal clips

Mounting Hardware:
2 – 1.5 in. P-style clamps
12 in. Of Steel banding with 0.25 in. Holes
10 – 4in. zip ties
6 – 0.25in. x 20 1in. Long Red Robertson screws
4 – 0.25 in. Washers
6 – 0.25in. hex nuts
4 – M3 0.75in. hex head screws
4 – M3 hex nuts
4 - #8 0.75in. Phillips screws
4 - #8 hex nuts

Instruction set

Part 1: Setting up the trailer (See Table 1 below for figures)

Step 1: Set your trailer out, drill out 4 holes in the middle of the frame of the trailer, 6 inches down, with your 0.125in. bit. (See figure 1A)

Step 2: Get your two 4.3in. stainless steel plates and drill 2 0.125in. holes in each plate, one on each end of the plate.

Step 3: Mount the plates by lining up the holes in the trailer with the holes in the plates (this will set our angle for the launcher) and mount them using your #8 hardware and Phillips screwdriver. (See figure 1B)

Step 4: Get your multi-tool and measure 1 inch from the back of the trailer in the center. Cut a centered 1.1in. x 0.6in. hole in the bottom of the trailer. (see figure 1C)

Step 5: Drill 2 0.125in. holes on either side of the hole (See figure 1C)

Step 6: still using the multitool, measure 1.6 inches from the front of the trailer, and on the left side, cut a 0.5in. x 1in. Hole in the bottom of the trailer (See figure 1F)

Step 6: Measure out and drill holes with your 0.125in. bit as follows on the left wall (See figure 1D)

- 1.4-inch back, 0.5-inch down
- 3.5-inch back, 2.6-inch down
- 6.2-inch back, 0.5-inch down
- 4.2-inch back, 2.6-inch down.

Step 7: Get your 0.25in. bit, and on the front of the trailer, measure out from the center and drill:

- 1 hole at 1.5 inches right from the center. O.5 inch up
- 1 hole at 1.5 inches left from the center. O.5 inch down

(See figure 1E)

Step 8: Back on the bottom of the trailer, measure out from the front and drill: (See figure 1F)

- 1 hole at 0.5 inch back, 1 inch left from the center
- 1 hole at 0.5 inch back, 1 inch right from the center
- 1 hole at 5 inches back, 1 inch left from the center
- 1 hole at 5 inches back, 1 inch right from the center

Step 9: Use your multi-tool and cut a 1.7in. hole in the center of the front of the trailer (See figure 1E)

(FIGURES FOR PART 1 ON NEXT PAGE)

TABLE 1

		
Figure 1A	**Figure 1B**	**Figure 1C**
		
Figure 1D	**Figure 1E**	**Figure 1F**

Part 2: Making the Barrel (see Table 2 for figures)

Step 1: Cut your two feet of pipe into a 6-inch, 5-inch, and 4-inch piece (See figure 2A)

Step 2: Using your 3 ways, mount the 5-inch piece on the top, and the 4 and 6 on either side (doesn't matter) (See figure 2B)

Step 3: Using your multi-tool, measure 1-inch down the 6-inch piece and cut a 0.5 x 1 inch hole on the top and bottom of the cylinder. (See figure 2C)

TABLE 2

		
Figure 2A	**Figure 2B**	**Figure 2C**

Part 3: Reload system (see Table 3 for figures)

Step 1: Drill a 0.125in hole in one end of the 4 Inch steel (refer to figure 3A)

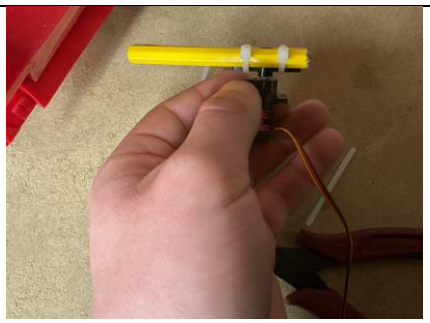
Step 2: Use two zip ties to secure the steel to the servo motor (see figure 3B)

Step 3: Take your plastic cylinder, drill a 0.25in hole a half inch deep in one end, and a 0.125in. hole through the diameter of the cylinder until it mates with the center hole.
(See figures 3C and 3D)

Step 4: Use your pliers to bend a half inch at either end of the brass rod in opposite directions. (See figure 3E)

Step 5: Slide one end of the rod into the cylinder and one end into the hole in the 4in. Plate. (See figure 3F

TABLE 3

		
Figure 3A	**Figure 3B**	**Figure 3C**
		
Figure 3D	**Figure 3E**	**Figure 3F**

Part 4: Mounting (See Table 4 below for figures)

Step 1: Slide your barrel through the hole in the front of the trailer.

Step 2: Use 4 0.25x 20 screws and nuts, as well as your metal strapping, to secure the barrel in place.
(see figure 4 F)

Step 3: Set your cylinder inside the barrel and set the servo motor of the reload system in the hole in the back of the trailer, secure using 2 zip ties. (see figures 4C and 4D)

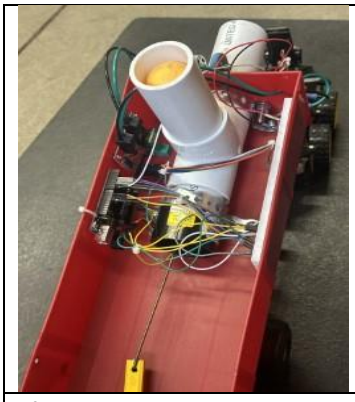
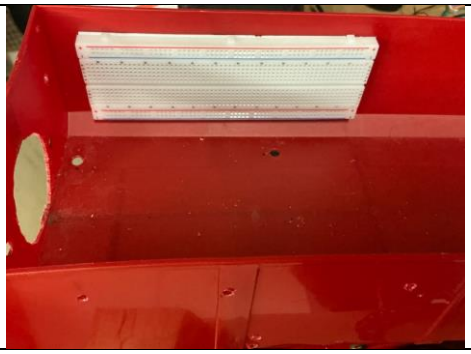
Step 4: Using the P-Clamps, 4 washers, and the other two 0.25x20 bolts and nuts to secure the flywheels and motors to the front of the trailer. (see figures 4E and 4F)

Step 5: Use the adhesive of the breadboard to mount it to the inside wall of the trailer.
(see figure 4 H)

Step 6: Use M3 nuts and hex screws to mount the power supply and Arduino boards.
(see figures 4A and 4B)

(FIGURES FOR PART 4 ON NEXT PAGE)

TABLE 4

		
Figure 4A	**Figure 4B**	**Figure 4C**
		
Figure 4D	**Figure 4E**	**Figure 4F**
		
Figure 4G	**Figure 4H**	

Part 5: Wiring (see Table 5 below for figures)

Step 1: Connect your battery to the power supply module using two 50cm jumpers. (refer to figures 5E and 5F respectively for visual)

Step 2: Cut your 16-G wires in half, strip them back, and use your terminal driver to mount the wires into the outputs of the power supply module. (see figures 5H for wires, and 5C for connections)

Step 3: Strip back the other ends of your 16-G wire, crimp the terminal clips on, and connect them to the flywheel motors. (see figure 5G for motor connections)

Step 5: Connect 2 50cm wires to the breadboard's positive and negative, and a third 50cm wire to the smartboard's "pin 7" port. Connect them to the 5V, ground, and pin 11 on the Elegoo car, respectively. This will power 5V to the breadboard and connect the smartboards to send signals. (see figure 5D)

Step 6: To connect your servo motor to power, attach a male-male wire to all 3 terminals in its receiver. The wire that is in the red goes to positive, the brown to ground, and the yellow to the pin 9 port on the smart board. This will draw off the car battery to run the servo. (See figures 5D and 5I)

Step 7: Connect the Vin and ground ports on the smart board to the breadboards positive and negative respectively. This sends 5V to the breadboard to operate. (see figure 5A and 5D)

Step 8: You will need 7 50cm male-female wires. Find the male receivers on the power supply module. Connect the ground pin to the ground on the breadboard. Connect "ENA" on the module to pin 11, and the other to pin 10 on the smartboard (see figures 5B and 5D)

Step 9: Connect INT 1 to pin 6. Connect INT 2 to Pin 5. Connect INT 3 to Pin 4. Connect INT 4 to Pin 3
(See figure 5B)

Step 10: Connect the 5V to the positive on the breadboard. (see figures 5A and 5D)

(FIGURES FOR PART 5 ON NEXT PAGE)